

Literature Status: Dimension-Free Convexity of Fermionic Entropy

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Status

This is a **literature-already-answered** packet. It records a full positive resolution, already present in later literature, of the all-dimensional Hessian and geodesic-convexity conjecture from arXiv:1203.5377. It is not a new proof packet.

Source Question

Carlen and Maas construct a non-commutative Riemannian transport metric for the fermionic Fokker–Planck equation in arXiv:1203.5377 [1]. Remark 5.6, on PDF page 24, proves for Clifford dimensions $n = 1, 2$ that

$$\text{Hess}_\rho S(\nabla U, \nabla U) \geq \|\nabla U\|_\rho^2,$$

and conjectures that this inequality holds for every n .

Supporting Answer

The introduction of arXiv:1609.01254, on PDF page 2, explicitly says that the paper proves the sharp geodesic-convexity result for the Fermi Ornstein–Uhlenbeck semigroup and thereby solves the problem left open in [1]. This satisfies the explicit-awareness criterion for the literature-already-answered classification.

The decisive statement is Theorem 8.6 on PDF page 34 [2]. For inverse temperature $\beta \geq 0$, it proves that relative entropy is geodesically λ_β -convex in the Riemannian metric associated with the Fermi Ornstein–Uhlenbeck generator, where

$$\lambda_\beta = \min_j \cosh(\beta e_j/2).$$

The paragraph immediately preceding that theorem explains that the metric can be written using the skew derivations exactly as in [1]. At $\beta = 0$, the invariant state is tracial and $\lambda_0 = 1$. Thus the theorem gives geodesic 1-convexity for the source metric in every Clifford dimension. In this smooth finite-dimensional Riemannian setting, this is equivalent to

$$\text{Hess}_\rho S \geq g_\rho,$$

which is precisely the conjectured inequality.

Scope

The Hessian and geodesic-convexity conjecture is fully answered positively, with the sharp constant 1. The source paper’s separate Remark 5.7 question about the sectional curvature of the density manifold is not answered by this identification.

Search Evidence

The run’s cheap indexes were searched for arXiv:1203.5377 and the terms “fermionic”, “Hessian”, “entropy convexity”, “Ricci”, and “Clifford”; no prior exact packet was present. A bounded local-source and arXiv search located arXiv:1609.01254. Its introduction explicitly identifies the earlier problem as solved, and Theorem 8.6 supplies the exact all-dimensional sharp statement.

References

- [1] E. A. Carlen and J. Maas, *An Analog of the 2-Wasserstein Metric in Non-commutative Probability under which the Fermionic Fokker-Planck Equation is Gradient Flow for the Entropy*, arXiv:1203.5377; Comm. Math. Phys. 331 (2014), 887–926.
- [2] E. A. Carlen and J. Maas, *Gradient Flow and Entropy Inequalities for Quantum Markov Semigroups with Detailed Balance*, arXiv:1609.01254; J. Funct. Anal. 273 (2017), 1810–1869.